Ethers

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Contents

1 Ether Synthesis

Core Concept 1.1: Ethers

Ethers are a class of organic compounds characterized by an oxygen bonded to two alkyl or aryl groups. Their general formula is R—O—R', where $R = R'$ (symmetrical ether) or $R \neq R'$ (asymmetrical ether).

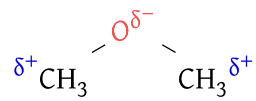

Figure 1: Dimethyl ether structure

Core Concept 1.2: Williamson Ether Synthesis

Williamson ether synthesis is the primary way chemists make asymmetrical ethers. This method uses an alkoxide $R'O^-$ as its nucleophile, thus works best when attacking non-β-branched primary R – Xs because β-branched, secondary, or tertiary R – Xs will unintendedly use the E2 mechanism.

$$R_{prim} - X + R'O^- \xrightarrow{S_N2} R - O - R'$$

It's important to note that the if the desired product is asymmetrical, the choice of corresponding R groups of the haloalkane and alkoxdide is important because a tertiary or secondary haloalkane paired with a primary alkoxide will lead to E2 whereas the opposite combination will undergo S_N2 (*the desired method*).

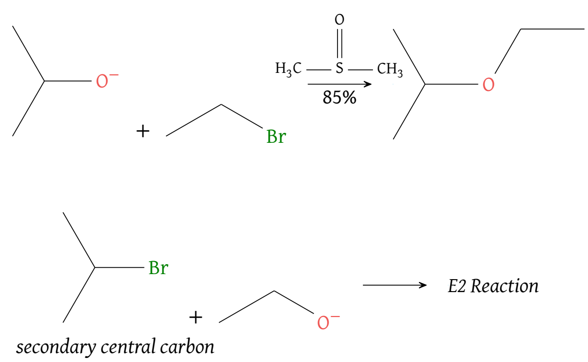

Figure 2: Williamson ether retrosynthesis; *unsymmetrical ethers have two choices, one is usually favorable over the other*

Core Concept 1.3: Alcoholysis of $R_{sec/tert}X$

For $R_{sec/tert}X$ haloalkanes, chemists use the protonated version of alkoxides (alcohol) to make the nucleophile less basic and consequently undergo S_N1 or E1 rather than E2.

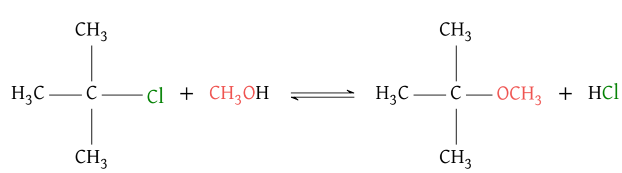

Figure 3: Alcoholysis of $R_{sec/tert}X$

While E1 reactions still remain an unintended pathway, chemists minimize this by carrying out the reaction in lower temperatures.

Core Concept 1.4: Intramolecular Williamson Ether Synthesis (*Cyclic Ether Formation*)

So far, we've looked at only **intermolecular** ether synthesis reactions. **Intramolecular** ether synthesis (*or cyclic ether formation* works in the same way as the intermolecular equivalent does.

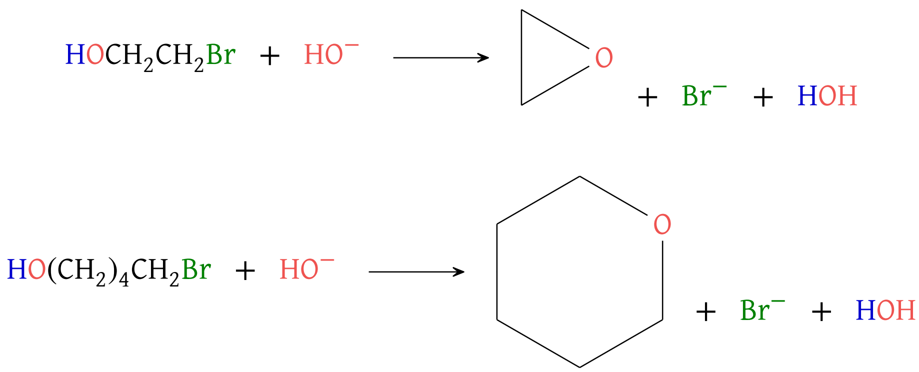

Figure 4: Intramolecular Williamson ether synthesis

Intramolecular are particularly favorable over intermolecular ether synthesis because [1] it does not lower entropy and [2] highly diluted concentrations of the relevant alcohol do not effect it as much because the reaction is occurring with itself.

Mechanistically, this reaction occurs via a backside attack with inversion, prepared by the rearrangement of a less stable rotamer if needed.

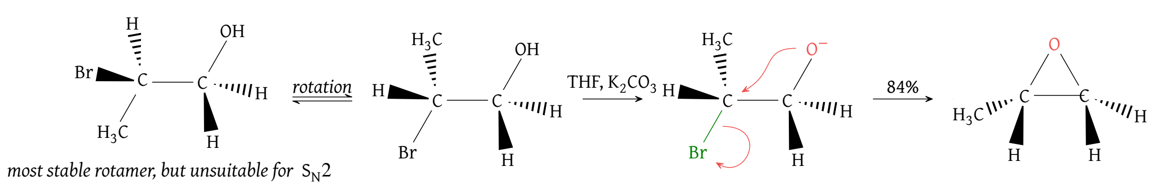

Figure 5: Backside attack and inversion in cyclic ether formation

The relative rates of ring closure depend on the trade off between ΔH (*ring strain*) and ΔS (*lining up*) in the various transition states. The following is the ranking of ether ring formation speeds by ring size.

$$3 > 5 \geq 6 > 4 \geq 7 > 8$$

Reaction speed does not directly correspond to ring size. Although one might expect a linear relationship, the trend is more nuanced: 3-membered rings react fastest owing to the close proximity of the bonding atoms, 4-membered rings are the slowest, and 6-membered rings are surprisingly fast again due to favorable atomic geometry. In summary, this ordering stems from competing ΔH and ΔS contributions across the transition states of each ring size, rather than any simple geometric trend.

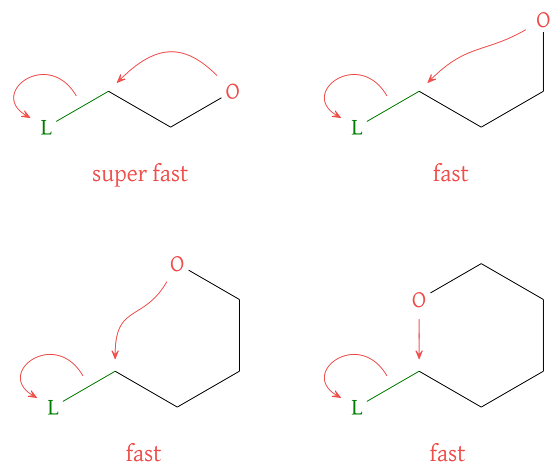

Figure 6: Relative rates of ring closure

Core Concept 1.5: Symmetrical Ether Synthesis via Alcohols

Ethers can also be formed by **alcohols and H^+ through** S_N2 **and/or** S_N1 depending on the R groups. Chemists specifically use this technique to form symmetrical ethers because and alcohol mixure would otherwise produce a slurry of different ether product.

$$2\,R-OH \xrightarrow[-H_2O]{H_2SO_4} R-O-R$$

For $R_{prim}-OH$, this mechanism works via S_N2 due to minimal steric hinderance.

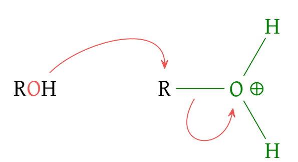

Figure 7: Symmetrical ether formation via S_N2

When dealing with $R_{sec/tert}-OH$, this mechanism shifts to S_N1 .

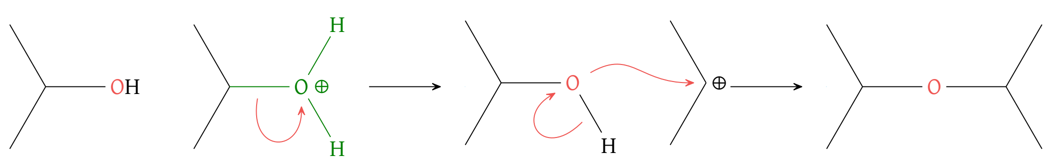

Figure 8: Symmetrical ether formation via S_N1

Core Concept 1.6: Asymmetrical Ether Synthesis (*Tetrabutyl Scenario*)

However, a unique alcohol combination that preferentially forms an asymetrical ether is an **unhindered ROH and tetrabutyl mixture**.

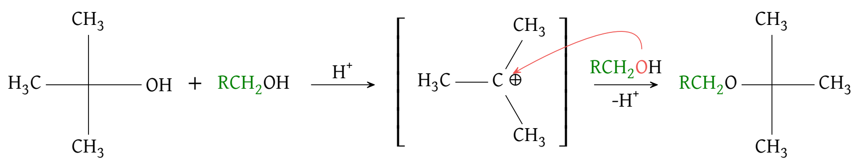

Figure 9: Asymmetrical ether formation with t-Butyl group

The ROH preferentially attacks readily avaliable t-Bu cations (*due to hyperconjugation*) via S_N1. And while the t-Bu cation could be attacked by another t-Bu alcohol molecule, steric hinderance makes this interaction less likely to occur compared to the smaller, less sterically hindered ROH.

In the laboratory, t-Bu is used for **"protection"** of other alcohols by forming these t-Bu ethers.

2 Ether Reactions

Let's go through a collection of different ether reactions.

Core Concept 2.1: $R_{prim}-O-R_{prim}$ Reaction

$R_{prim}-O-R_{prim}$ are quite unreactive, exhibiting stability to bases, RLi, RMgBr, and dilute aqueous H^+. However they are reactive under H^+X^- exposure.

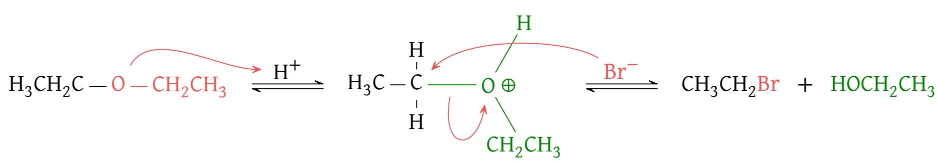

Figure 10: Ether cleavage with hydrohalic acids

Core Concept 2.2: $R_{prim}-O-R_{sec}$ Reaction

$R_{prim}-O-R_{sec}$ ethers react either through S_N1 or S_N2 depending on the nucleophile strength (*under acidic conditions*). When paired with a **good nucleophile**, the attack occurs on the R_{prim} side carbon via S_N2 to avoid the steric hinderance of R_{sec}.

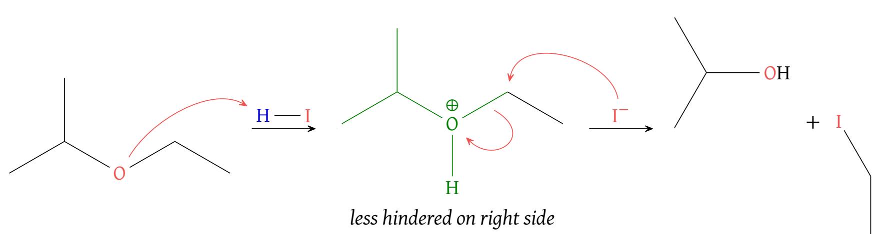

Figure 11: Mixed ether reaction with good nucleophile (S_N2)

When paired with a **poor nucleophile**, an S_N1 reaction occurs at the carbocation of R_{sec} (*stabalized by hyperconjugation*).

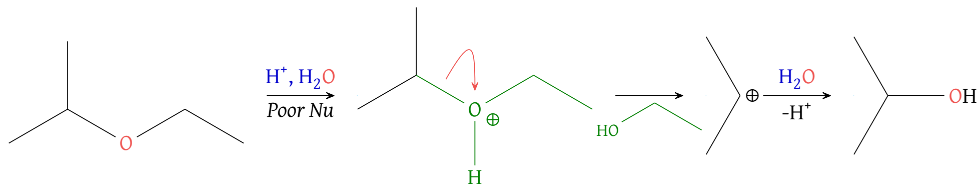

Figure 12: Mixed ether reaction with poor nucleophile (S_N1)

Core Concept 2.3: t-Bu Ether Hydrolysis

t-Bu ether hydrolysis is the reverse of t-Bu ether formation covered in Core Concept 1.6. This reverse process is the "deprotection" of alcohol ROH by the t-Bu cation via acid. The resulting t-Bu cation, when exposed to heat, undergoes E1 to exaporate into an alkene.

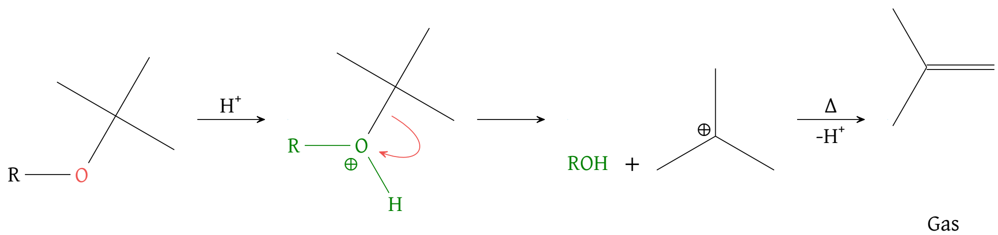

Figure 13: Hydrolysis and deprotection of t-Butyl ethers

Core Concept 2.4: Ether Ring Opening

Ether ring openings release ring strain by a Nu^- attacking directly one of the two carbons adjacent to the oxygen. This is a regioselective process, meaning S_N2 occurs at the less hindered relevant carbon.

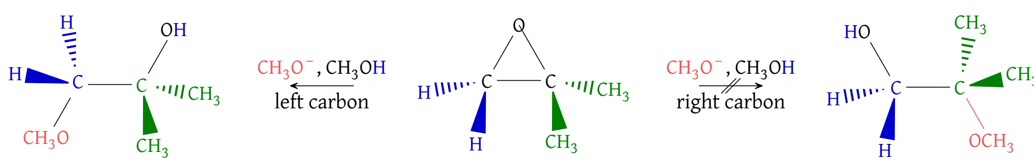

Figure 14: Regioselective ether ring opening

Many Nu^- will work. The following example shows that ether ring openings are regioselective and also **stereocontrolled**.

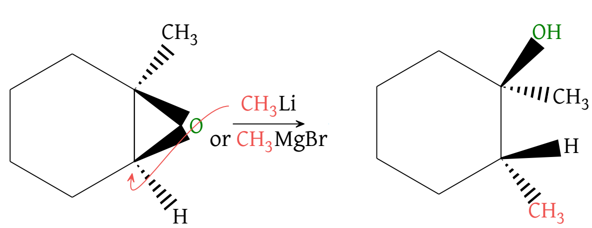

Figure 15: Stereocontrolled ether ring opening

With **neutral nucleophiles**, acidic conditions are required to activate the ether toward nucleophilic attack. Without H^+, the reaction will not occur.

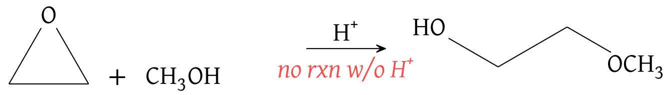

Figure 16: Acid-activated ether ring opening with neutral nucleophile

One important distinction in the neutral nucleophile mechanism is regioselectivity towards the more hindered carbon. In asymmetrical ethers, the nucleophile preferentially attacks the more substituted carbon, as it better stabilizes the partial positive charge (δ^+) that develops through hyperconjugation and inductive effects—causing the C-O bond on that side to stretch and weaken. This gives the mechanism a hybrid S_N1 / S_N2 character: it resembles S_N2 in that nucleophile attack and leaving group departure occur simultaneously, yet resembles S_N1 in that the partial positive charge mimics a carbocation.

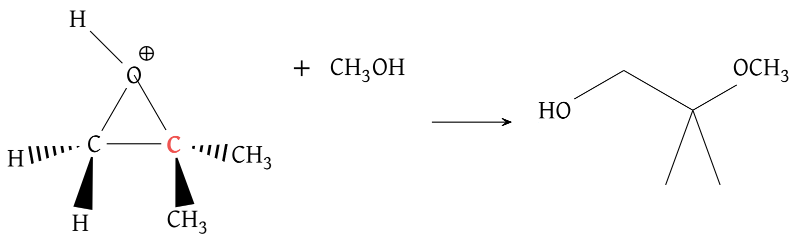

Figure 17: Regioselectivity of acid-catalyed ring openign of asymmetrical ethers

3 Sulfur Analogs: *A Brief Lesson*

Core Concept 3.1: Sulfur Analogs

Sulfur is directly below oxygen in the periodic table, and thus behaves very similarly.

Thiols (RSH) are **more acidic than alcohols** because the S–H bond is weaker due to sulfur's higher-energy molecular orbitals. The lower electronegativity difference in S–H compared to O–H also results in **weaker hydrogen bonding**, making thiols less polar overall. *This is illustrated by H_2S, which is a gas at 25°C with a boiling point of* -60*°C, in stark contrast to H_2O.*

Despite being less basic than RO^-, RS^- is a stronger nucleophile due to its greater **polarizability**—note that polarizability is distinct from polarity. Polarizability describes how easily an electron cloud can be distorted by an external force (*Core Concept 2.4*). Larger atoms like sulfur are more polarizable than oxygen because their diffuse, loosely-held electron clouds are more easily deformed. This allows RS^- to stretch and distort toward an electrophile, enhancing its nucleophilicity despite its lower basicity.

Core Concept 3.2: Sulfur Reactivity

The better nucleophility of RS^- eliminates the issue of E2 reactions when paired with $R_{sec}X$ like RO^- has.

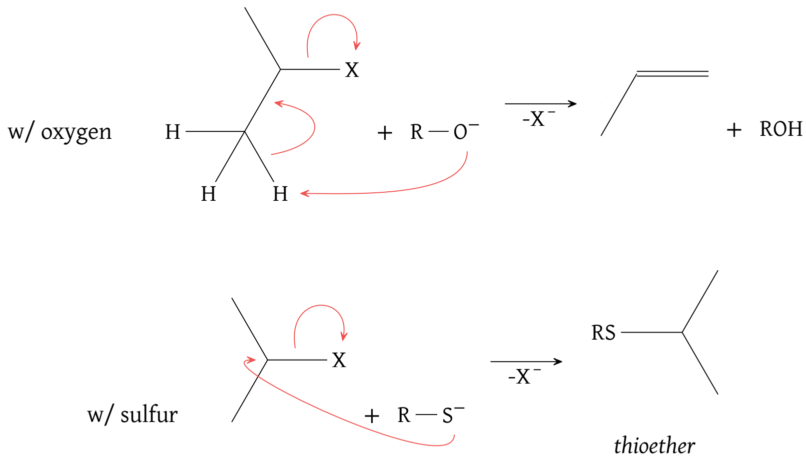

Figure 18: Sulfur versus oxygen reaction with $R_{sec}X$

Even neutral RSR’ readily undergoes S_N2 , whereas dimethyl ether (CH_3OCH_3) does not react under the same conditions. When RSR’ attacks an RX, the resulting sulfonium ion (R_3S^+) becomes susceptible to attack by another nucleophile — the sulfide, once a good nucleophile, has now become a good leaving group, analogous to H_2O. This makes **sulfonium salts** (R_3S^+) effective **alkylating agents**, meaning they can transfer an alkyl group to an incoming nucleophile.

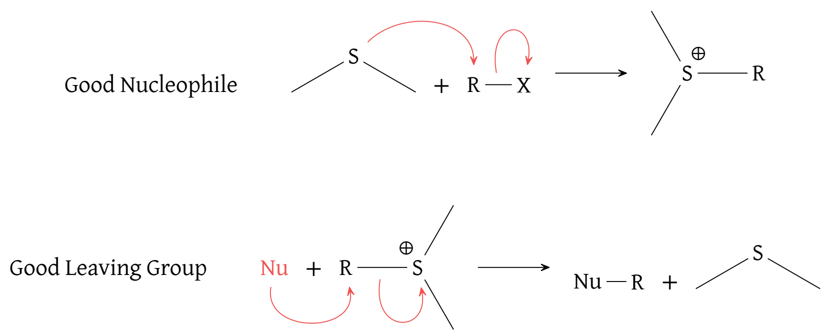

Figure 19: Sulfide acting as a nucleophile to form a sulfonium ion (*top*), which then acts as an alkylating agent toward an incoming nucleophile (*bottom*)